MILITARY SPECIFICATION

BATTERY, STORAGE, AIRCRAFT, TYPE MA-2-1

1. SCOPE

1.1 This specification covers one type of 19-cell, 24 volt, nickel-cadmium battery, designated type MA-2-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-C-502	Copper Rods And Shapes; And Flat Products With Finished Edges (Flat Wire, Strips And Bars)
QQ-C-576	Copper Flat Products With Slit, Slit And Edge-Rolled, Sheared, Sawed, Or Machine Edges, (Plate, Bar, Sheet And Strip)
QQ-S-766	Steel Plates, Sheets, And Strip-Corrosion Resisting
PPP-F-320	Fiberboard; Corrugated And Solid, Sheet Stock (Container Grade), And Cut Shapes
PPP-B-601	Boxes, Wood, Cleated-Plywood

Military

MIL-D-1000	Drawings, Engineering And Associated Lists
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-P-6063	Packaging Of Batteries, Storage, Charged And Dry-Uncharged And Moist, General Specification For

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MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Specifications And Standards Order Of Precedence For The Selection Of
MIL-STD-831	Test Reports, Preparation Of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The battery furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids. (See 4.5 and 6.4)

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as specified in 3.2.1 and 3.2.2.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception specified in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials.

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3.3.1 Insulating materials and compounds. Insulating materials and compounds shall be free from any and all materials soluble in potassium-hydroxide solution.

3.3.2 Battery canister. Stainless steel conforming to QQ-S-766 and having a minimum thickness of 0.044 inch shall be used in the fabrication of the battery tray and cover.

3.3.2.1 Battery canister lining. The canister including the cover shall be suitably protected with epoxy base compound coating baked on to a thickness not to exceed 0.015 inch. This coating shall be so bonded to the canister material that it will not separate, sag, crack or chip when subjected to the tests specified herein.

3.3.3 Cell jar. Nonmetallic material conforming to requirements specified herein shall be used in the fabrication of cell jars.

3.3.4 Protective treatment. When materials are used in the construction of the battery that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 Design and construction. The requirements specified herein cover a battery consisting of two units. Unit No. 10 shall contain 10 cells conforming to the physical requirements of Figures 1 through 7. Unit No. 9 shall contain 9 cells conforming to the physical requirements of unit No. 10 except that the vent nozzle shall be on the same end as the negative terminal and the nameplate on the same end as the positive terminal. On unit No. 9 the space taken by the tenth cell on unit No. 10 shall be filled with a suitable alkaline resistant blocking material. Unit No. 9 shall have green paint stripes and unit No. 10 shall have red paint stripes. The battery shall be designed to permit easy removal of individual cell jars for replacement purposes.

3.4.1 Stability. The battery shall be so designed and constructed as to be electrically and thermally stable and shall not require the use of an electromagnetic protective device.

3.4.2 Cell-jar construction. Cell jars and cell-jar caps shall be so designed and constructed as to permit easy servicing with electrolyte. A plainly visible mark or line shall be incorporated in the cell jar to permit personnel to determine the correct electrolyte level. The method used for designating the electrolyte level shall be approved by the procuring activity. The dimensions of the cells shall conform to Figure 6.

3.4.2.1 Baffle. A nonmetallic baffle shall be so incorporated within the individual cell jars as to protect the plates and separators from objects inserted into the vent openings.

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3.4.3 Plates. The plates shall be free from imperfections which may affect their structural strength and performance. Active material shall be incorporated into the plates in accordance with high-grade commercial practice.

3.4.4 Separators. Separators shall be free from flaws and other imperfections which may adversely affect the performance of the battery.

3.4.5 Intercell connectors. The intercell connectors shall be constructed of copper in accordance with QQ-C-502 or QQ-C-576. They shall be of sufficient cross-section to withstand, without failure or excessive heating, any discharge rate specified herein. The intercell connectors shall be so protected as to be unaffected by a solution of potassium hydroxide.

3.4.6 Cell terminal posts. Cell terminal posts shall be constructed of material that is unaffected by a solution of potassium hydroxide. They shall conduct adequately any current specified herein.

3.4.7 Vents.

3.4.7.1 Cell vents. Each cell jar shall have a valve opening into the void space between the cell jar and the battery cover. This valve shall be designed to permit escape of battery gas at pressures of 6 ± 4 pounds per square inch (psi) and above. The valve shall prevent entry of air into the cell under a pressure of 12 psi.

3.4.7.2 Battery vents. Each battery tray shall be provided with a removable vent nozzle and removable vent plug as shown on Figure 1. This vent shall provide for free venting of gas and electrolyte spray from the void space between the cell jars and the battery cover. The dimensions of this vent nozzle shall be as shown on Figure 3.

3.4.8 Battery covers. Each battery tray shall be fitted with a removable cover to permit easy access to the individual cell jars. When in place, the cover shall so seal the void space above the cell jars as to maintain the internal gas pressure with a head of 55 inches of water for a period of 2 minutes without loss of monometer pressure in excess of 2.75 inches of water through other than the vent nozzles. This cover shall be made of corrosion-resistant steel. Dimensions of the cover shall conform to Figure 5.

3.4.9 Condition. Unless otherwise specified, the battery shall be furnished with electrolyte in a 70 to 95 percent charged condition. (See 6.2)

3.4.10 Battery terminals. The battery terminals shall be as shown on Figure 4 and installed on the battery as shown on Figure 1.

3.5 Performance.

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3.5.1 Environmental. The battery and all components thereof shall be constructed of material not permanently affected under the following conditions or natural combination of conditions:

a. Temperatures ranging from -63° to +68° Centigrade (C)

b. Temperature shock of +80°C high temperature.

c. Pressures ranging from 30 inches of mercury down to 3.4 inches of mercury (approximately an altitude of 50,000 feet).

d. Radiant energy as found under natural conditions.

e. Acceleration and shock forces as anticipated.

3.5.2 High-rate discharge. The cell terminal, intercell connectors, and terminal connectors of the battery shall be capable of withstanding, without evidence of physical damage, a discharge of 1,500 amperes (amp) for 30 seconds. (See 4.4.2)

3.5.3 Insulation. The battery shall be capable of withstanding a voltage of 2,500 volts (v) root mean square (rms) at commercial frequency for 1 minute between current-carrying parts and the case without breakdown of insulation. (See 4.4.3)

3.5.4 Life. The battery shall have a minimum cycle life of 400 charge-discharge cycles as specified in 4.4.4.

3.5.4.1 The discharge capacities and cutoff voltages of the battery at the various required discharge rates shall be as shown in Table I. Conformance of the battery to these requirements shall be determined during the initial discharge characteristic cycles specified in 4.4.4. The battery shall not develop less than 100 percent rated capacity at each of the specified rates during the initial discharge characteristic cycles.

TABLE I

Rate of Discharge	Temp. °C ±3	5-second Voltage	End Voltage	Battery Capacity (See 6.3)	
				Time	AH
30	27	--	20.0	2 hrs	60
500	27	18.5	15.0	4 min	33.33
1,000	27	15.0	10.0	1.5 min	25.00
500	-18	15.0	15.0	2.5 min	20.83

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3.5.5 Gas tightness of cell jars. The cell jars shall be capable of withstanding an internal air pressure of 10 psi (or the maximum design pressure of the valve) and an external air pressure of 13 psi for 5 minutes without leakage. (See 4.4.5)

3.5.6 Charge acceptance. The battery shall satisfactorily charge on a constant potential of 28.5v when charged and discharged in accordance with 4.4.6.

3.5.7 Retention of charge. The battery shall maintain charge when tested in accordance with 4.4.7.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-D-1000 shall govern changes in the manufacturer's part numbers.

3.7 Screw threads. Screw threads shall be in accordance with MIL-S-7742.

3.8 Weight. When filled with electrolyte to the normal level, the weight of the battery shall be not more than 160 pounds.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. Units No. 9 and 10 each shall be clearly identified on the nameplate in the location shown on Figure 1 The nameplate shall include the following information:

Nominal Capacity: 60 AH at 2-Hour Rate
Open Circuit Voltage: * Volts **
Unit No. *, * Cells **
Charging Voltage: 28.5v; units 9 and 10 combined
Serial Number * Type No. MA-2-1
Manufacturer's Name *
Contract Number *
Date of Manufacture (Month and Year)*

* To be filled in by the manufacturer.
** This will be different on units 9 and 10.

3.9.1 Additional marking. The following note shall be marked on each side of both battery units No. 9 and 10:

ONE BATTERY CONSISTS OF ONE RED STRIPE SECTION
AND ONE GREEN STRIPE SECTION.

3.10 Terminal marking. The polarity of the battery terminal posts shall be marked with a red (+) symbol for positive and a black (-) symbol for negative in the locations shown on Figure 1.

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3.11 Workmanship. The battery, including all parts, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp edges, accuracy of dimensions, marking of parts, et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the battery shall be classified as follows:

a. Acceptance tests — See 4.3

b. Qualification testing — See 4.5

4.3 Acceptance tests. Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling plan and tests

4.3.1 Individual tests. Each battery shall be subjected to the following tests as described under 4.4:

a. Examination of product

b. High-rate discharge

c. Insulation

d. Gas tightness test

4.3.2 Sampling plan and tests.

4.3.2.1 Lot. A lot shall consist of 500 batteries or fraction thereof manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.3.2.2 Sampling plan. Two batteries shall be selected at random from each lot or fraction thereof on the order and subjected to the following tests as described under 4.4:

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Battery A	Battery B
Examination of product	Examination of product
High-rate discharge	High-rate discharge
Insulation	Insulation
Charge acceptance	Life

4.3.2.2.1 Rejection and retest. When one or more batteries from a lot fail to meet the specification, acceptance of all batteries in the lot will be withheld until the extent and cause of failure are determined or until approved by the procuring activity. After corrections have been made, all necessary tests shall be repeated.

4.3.2.2.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. However, final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3.3 Defects in batteries already accepted. The investigation of a test failure could indicate that defects may exist in batteries already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test methods.

4.4.1 Examination of product. Each battery unit shall be examined to determine compliance with the requirements specified herein with respect to dimensions, materials, workmanship, and marking. A portion of this examination shall include inverting the battery unit for not less than 60 seconds. If leakage of electrolyte occurs upon inversion, the battery shall not be acceptable.

4.4.2 High-rate discharge. Each battery unit shall be subjected to a high-rate discharge test for 30 seconds. The test shall be so run that an average current of 1,500 amp is produced. For convenience in testing, the rate may be maintained by a booster battery or motor-generator set. The cell terminal posts, intercell connectors, and terminal connectors shall withstand this discharge without showing evidence of failure. Fusing of any parts or any sign of physical change shall be cause for rejection. (See 3.5.2)

4.4.3 Insulation. Each battery unit shall have a voltage of 2,500v rms at commercial frequency applied between the current-carrying parts and the battery tray for a period of not less than 1 minute. A female sleeve connector having an outside diameter of 0.875 +000 -.016 shall be fitted over each battery terminal and shall engage at least 1/2 inch of the terminal. A connector of this type shall be applied to both terminals of the battery unit. (See 3.5.3)

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4.4.4 Life. The cycle life test shall consist of cycles of charge and discharge as specified in Table II. Discharges shall be accomplished at constant current for the measured initial, intermediate, and final capacity discharges.

TABLE II

LIFE CYCLE TEST

Cycle No.	Charging Discharging
Initial	In accordance with manufacturer's instructions
1 through 4	The battery shall be subjected to and shall pass all tests required to meet Table I requirements. Recharge shall be at a 20 to 30 amp rate. (See Notes 1 through 3)
5 through 100	Automatic cycling. (See Note 4)
101 through 104	Repeat Table I tests. The battery shall be capable of 90 percent of Table I requirements.
105 through 200	Automatic cycling.
201 through 204	Repeat Table I tests. The battery shall be capable of 80 percent of Table I requirements.
205 through 300	Automatic cycling.
301 through 304	Repeat Table I tests. The battery shall be capable of 80 percent of Table I requirements.
305 through 396	Automatic cycling.
397 through 400	Repeat Table I tests. The battery shall be capable of 80 percent of Table I requirements.

NOTE 1. The recharge period shall be of sufficient duration to charge the battery with 140 percent of the ampere-hours removed on the previous discharge. All low-temperature discharges shall be preceded by a charge at that temperature. The battery shall be soaked at that temperature for a period of at least 24 hours prior to the charge.

NOTE 2. Capacity tests shall be run at an initial electrolyte temperature specified in Table I, and from 2 to 80 hours after end of charge.

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NOTE 3. If the battery fails to meet the capacity requirements of Table I, as many as four repeat tests shall be permitted to demonstrate conformance to the capacity requirements specified in Table I. Repeat tests shall not be counted as life cycles.

NOTE 4. One automatic cycle will be 4 hours in duration. Six cycles shall be run in a 24-hour period. A minimum of 20 cycles shall be run in a 7-day period. The following schedule shall be followed:

Discharge 100 amp for 60 seconds

Discharge 600 amp for 30 seconds

Discharge 100 amp for 60 seconds

Discharge 600 amp for 30 seconds

Discharge 100 amp for 60 seconds

Discharge 600 amp for 30 seconds

Rest 0 to 3 minutes

Charge 28.5v at battery terminals - constant potential 1 1/4 hours

Rest for remainder of 4 hours, then repeat. The discharges shall be made into constant resistances of such value that the average current specified is maintained. The proper resistance values shall be determined during the first 20 cycles, then left constant for the remainder of the test. If desired, the battery may be immersed in a water bath of 21° to 32°C for the automatic cycling, or cooled by means of a fan. Distilled water may be added in accordance with manufacturer's instructions.

4.4.5 Vibration and gas tightness of cell jars. One battery shall be placed in an oven at 68°C and maintained at this temperature for at least 6 hours. The battery shall be allowed to cool to 40°C and an internal air pressure of 10 psi (or the maximum design pressure of the valve) shall be applied to the individual cell jars and maintained constant for 5 minutes. Following the internal pressure test, an external air pressure of 13 psi shall be applied to the cell jars for a period of 5 minutes. No leakage shall occur as a result of either test. Upon the successful completion of this part of the test, the battery shall be cooled to -34°C with all cells filled with electrolyte of the required specific gravity. The battery in normally mounted position shall be vibrated in accordance with procedure XI of MIL-E-5272. The temperature of the battery at start shall be -34°C and allowed to warm during the vibration period in an ambient temperature of 21° to 32°C. The battery shall be discharged at approximately the 2-hour rate for 30 seconds every 15 minutes. Voltage shall be recorded on a high-speed recording meter. Any discontinuities in voltage shall be cause for rejection. At the end of this time, visual inspection shall show no signs of cell fracture nor shall

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a repetition of the air pressure tests at 40°C reveal any leak from the cell terminal seal or any other part of the cell. For convenience in testing, each battery unit may be vibrated separately. (See 3.5.5)

4.4.6 Charge acceptance. One battery shall be fully charged and placed in an ambient temperature of 38° ± 2°C and allowed to remain for a period of not less than 16 hours. The battery shall then have a constant voltage of 30v direct current (dc) impressed across its terminals in the 38°C ambient. At the end of 8 hours, the current shall be not more than 5 amp. Following this test, the battery shall be placed in an ambient temperature of 24° to 30°C and allowed to cool for a period of at least 24 hours. The battery shall then be discharged at the 2-hour rate until the terminal voltage reaches 20v. A constant voltage of 28.5v shall be applied to the battery terminals for a period of 60 minutes. Following this charge and a 2-hour stand period, the battery shall be discharged at the 2-hour rate. The battery shall develop no less than 100 percent of its rated capacity. (See 3.5.6)

4.4.7 Retention of charge. One battery shall be subjected to, and shall pass, the requirements of the first 204 cycles specified in Table II; it shall then be fully charged by the constant-current method and stored in a location having an average temperature of not less than 20°C and not more than 25°C and an average relative humidity between 10 and 80 percent. After storage for not less than 60 days, the battery shall be discharged at the 500-amp rate. The capacity shall be no less than 65 percent of the capacity stated in Table I. (See 3.5.7)

4.4.8 Materials test. Samples of rubber or other elastomeric compounds used in the construction of the battery, along with samples of intercell connectors and cell terminal posts, shall be immersed in a solution of potassium hydroxide of 1.3 specific gravity at a temperature of 71°C for a period of not less than 4 weeks. After cleaning with water, no evidence of deterioration shall be visible.

4.4.9 Environmental tests. The battery shall be subjected to the following tests in accordance with the specified procedures of MIL-E-5272. Upon completion of the tests no sagging, cracking, crazing, or blistering of cell or battery cases or case lining shall occur.

4.4.9.1 High temperature. The battery shall be subjected to high temperature in accordance with procedure I except that maximum temperature of the chamber shall be 50°C. After 46 hours of exposure, one automatic life cycle (consisting of a discharge followed by a charge) shall be run. After cooling to room temperature, a 2-hour rate capacity check shall be run to insure compliance with the requirements of Table I.

4.4.9.2 Low temperature. The battery shall be subjected to low temperature in accordance with procedure II, except that the chamber shall be raised to -40°C in lieu of -54°C for operational test. After 20 hours of cold soak at -40°C, one cycle of automatic life cycle shall be conducted at -40°C. The discharges shall be through resistance values identical to those used in 4.4.9.1. The battery shall then be raised to room temperature and a capacity

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check given to determine the 2-hour rate capacity.

4.4.9.3 Temperature shock. The battery shall be fully charged and subjected to temperature shock in accordance with procedure I, except that the high temperature shall be 80°C rather than 85°C. Upon visual inspection of the battery after this test, there shall be no evidence of failure of any part.

4.4.9.4 Altitude. The battery shall be fully charged and subjected to altitude tests in accordance with procedure II. The battery shall be subjected to a charging voltage of 28.5v during the entire testing period. It shall remain at the reduced pressure specified for a period of 1 hour. The pressure transition periods (atmospheric to 50,000 feet and vice versa) shall be accomplished in 10 minutes or less. No visible evidence of leakage of electrolyte from the battery case or battery venting system shall occur. No evidence of cell failure shall be apparent.

4.4.9.5 Sunshine. The battery shall be subjected to sunshine in accordance with procedure I for a period of 100 hours. The covers shall be on the battery trays during this test.

4.4.9.6 Mechanical shock. The battery shall be subjected to shock in accordance with procedure II. The battery shall be discharging at approximately the 2-hour rate, and the discharge voltage shall be recorded by a high-speed recording voltmeter during the period of each shock test. Any discontinuity of the voltage resulting from a shock in any direction shall be cause for rejection.

4.4.9.7 Gas tightness test. The battery, with the cover secured in place, shall be subjected to a gas tightness test by connecting the vent nozzle through a manometer to a pressure equivalent to 55 inches of water. Any change in pressure greater than 2.75 inches of water during a 2-minute period shall be cause for rejection of the battery.

4.5 Qualification testing.

4.5.1 Qualification test samples tested by the contractor. The contractor shall subject four batteries and samples of rubber and other elastomeric compounds used in the construction of the battery, intercell connectors, and terminal posts to the qualification tests specified in 4.5.4.

4.5.2 Qualification test report. After the contractor completes the qualification tests, he shall prepare a qualification test report in accordance with MIL-STD-831.

4.5.3 Qualification test samples. The contractor shall submit samples, together with the test reports, to the Service Engineering Division. (See 6.5)

4.5.4 Qualification tests. Qualification tests shall consist of all tests specified in 4.4. Table III lists the tests to be conducted on each of the four samples and materials and the order in which they shall be run:

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TABLE III

Battery A	Battery B	Battery C	Battery D	Testing of Materials
Examination of Product	Examination of Product	Examination of Product	Examination of Product	Materials (See 4.4.8)
High-Rate Discharge	High-Rate Discharge	High-Rate Discharge	High-Rate Discharge	
Insulation	Insulation	Insulation	Insulation	
High Temperature	Retention of Charge	Life	Life	
Vibration and Gas Tightness	Charge Acceptance			
Mechanical Shock	Low Temperature			
Temperature Shock	Altitude			
Sunshine	Gas Tightness Test			

4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified. (See 6.2)

5.1.1 Level A. Each complete battery, composed of units 9 and 10, shall be packaged together in one unit container in accordance with MIL-P-6063. A minimum of 1 inch fiberboard pads conforming to PPP-F-320 shall be used between the two unit container. The unit container shall be the shipping container and shall conform to requirements of 5.2.

5.1.2 Level C. Each battery shall be preserved and packaged in a manner which affords adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. This level may conform to the supplier's commercial practice, provided the latter meets the requirements of this level.

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5.2 Packing. Packing shall be level A, B, or C, as specified. (See 6.2)

5.2.1 Level A. Each battery packaged as specified in 5.1.1 shall be packed in a container conforming to PPP-B-601, overseas type, style A or B. Closure shall be in accordance with the appendix of PPP-B-601.

5.2.2 Level B. Level B packing shall be the same as level A except that container shall be domestic type, PPP-B-601.

5.2.3 Level C. Each battery preserved and packaged as specified in 5.1.2 shall be packed in a manner which affords adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice, provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract or order the unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type MA-2-1 battery covered by this specification is intended for use in starting turbojet engines.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. When battery should be furnished with electrolyte other than in a 70 to 95 percent charged condition. (See 3.4.9)

c. Selection of applicable levels of preservation, packaging, and packing. (See Section 5)

6.3 Definitions.

6.3.1 Battery capacity. Battery capacity is the discharge measured quantitatively in ampere hours under specified conditions as to temperature, rate of discharge, and final voltage.

6.3.2 Cell jar or jar. The physical container housing the cell.

6.3.3 Battery tray or tray. The physical container housing the cell jars.

6.3.4 Battery unit or section. A tray containing the required number of cells.

6.3.5 Battery. The combination of two battery units, having a total of nineteen cells.

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6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: SANEOB, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

6.5 Test samples. Test samples shall be submitted to the San Antonio Air Materiel Area, Service Engineering Division, ATTN: SANEE, Kelly AFB, Texas 78241.

6.6 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

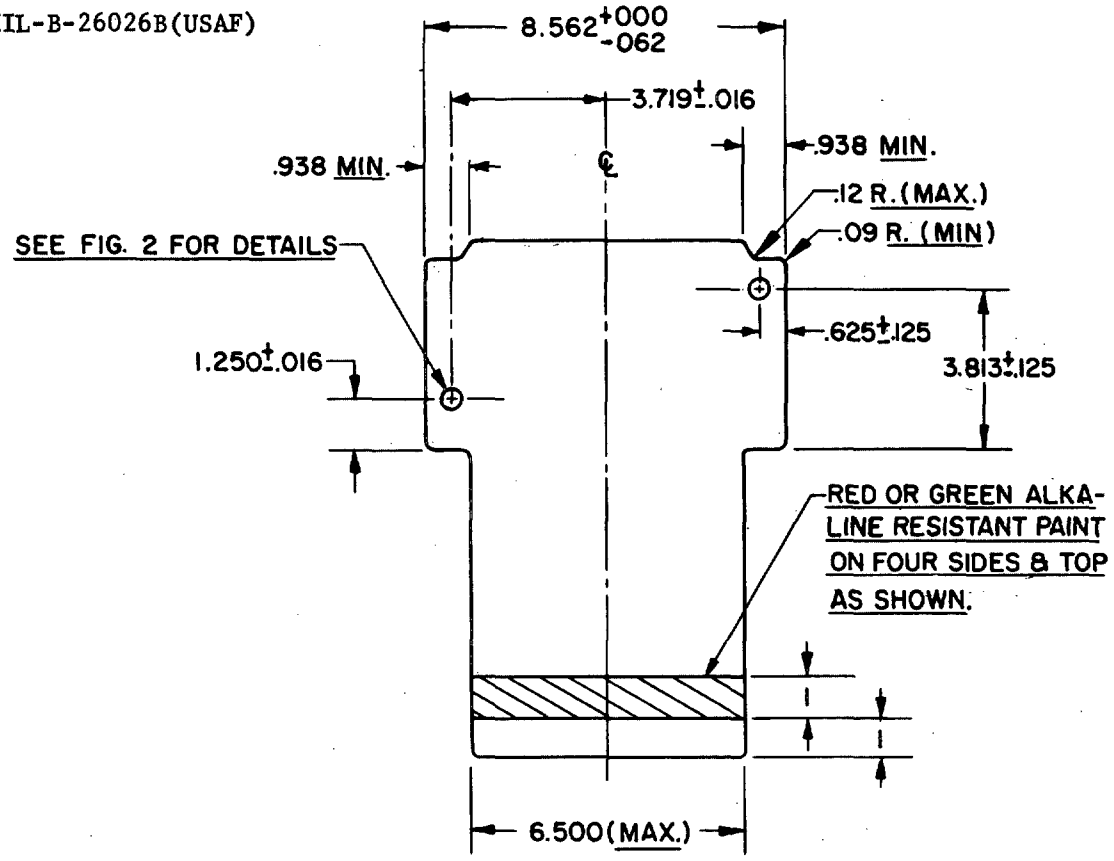
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Review Activities:
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Preparing Activity:
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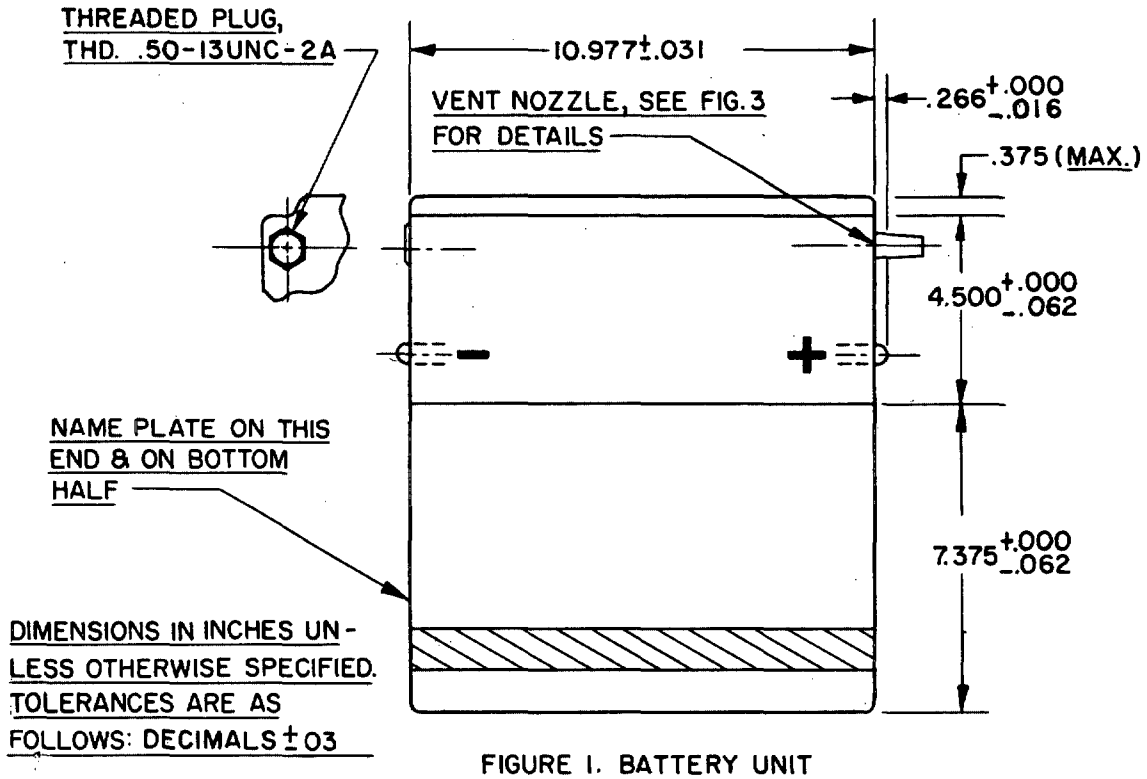
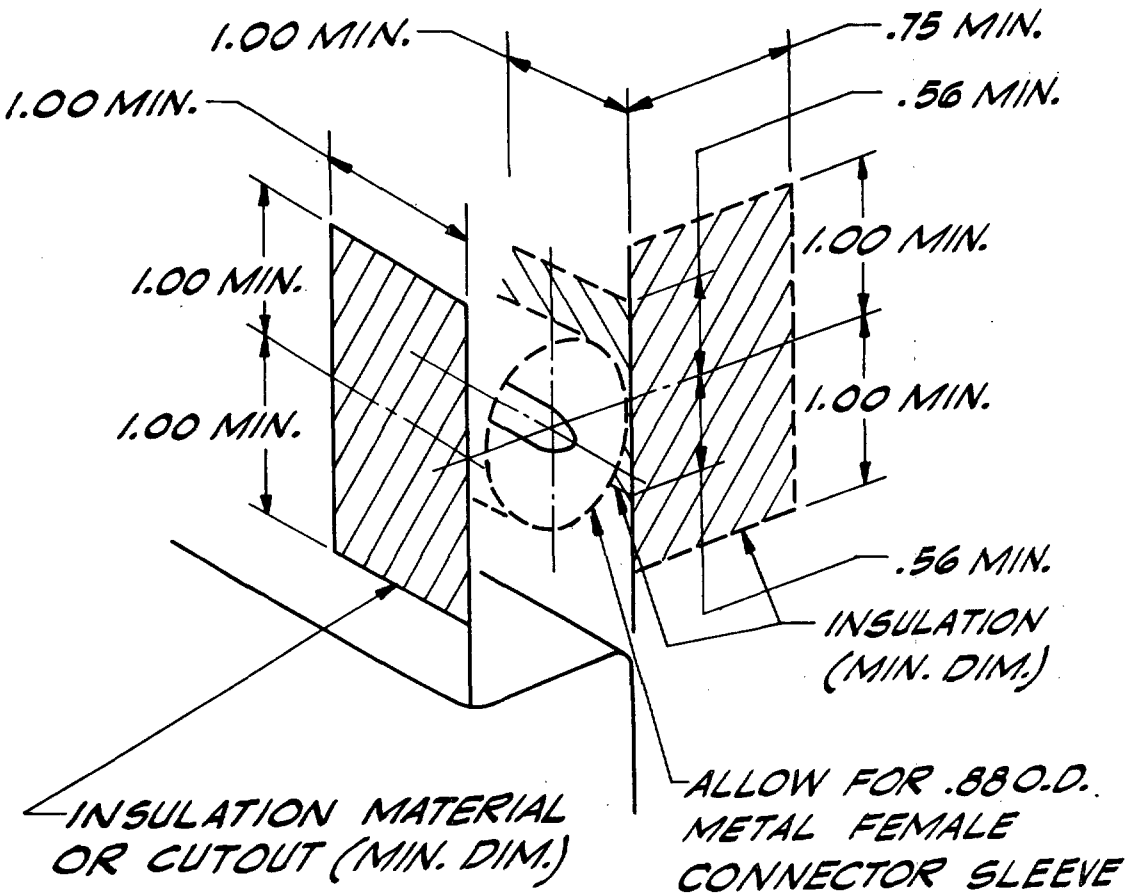


FIGURE 1. BATTERY UNIT

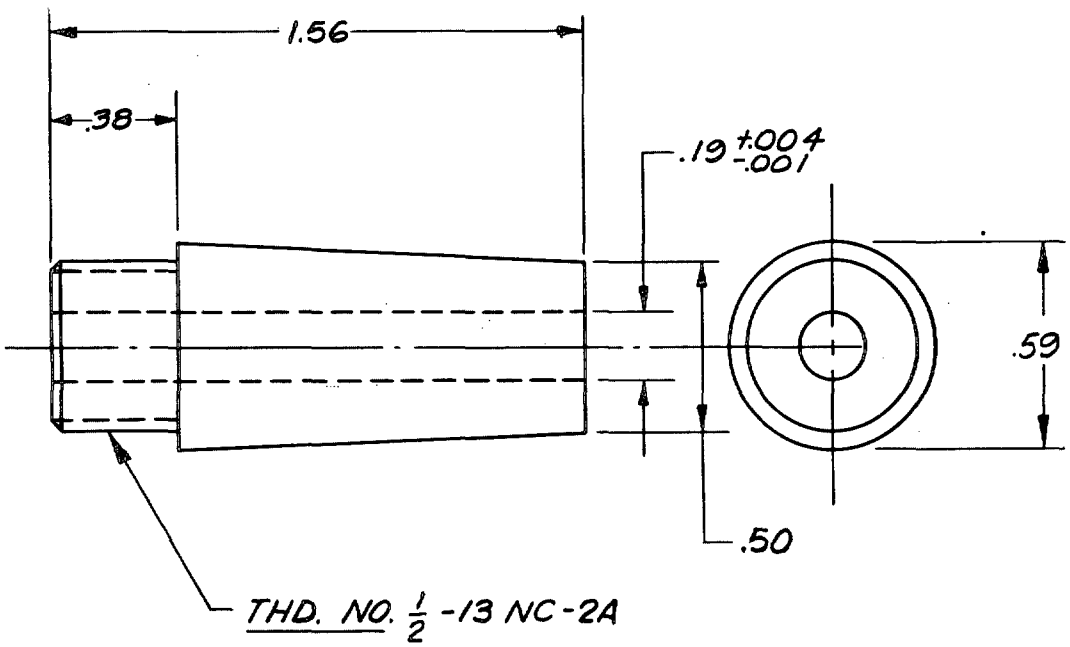
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DIMENSIONS IN INCHES

FIGURE 2 POSITIVE AND NEGATIVE TERMINAL INSULATION DETAILS

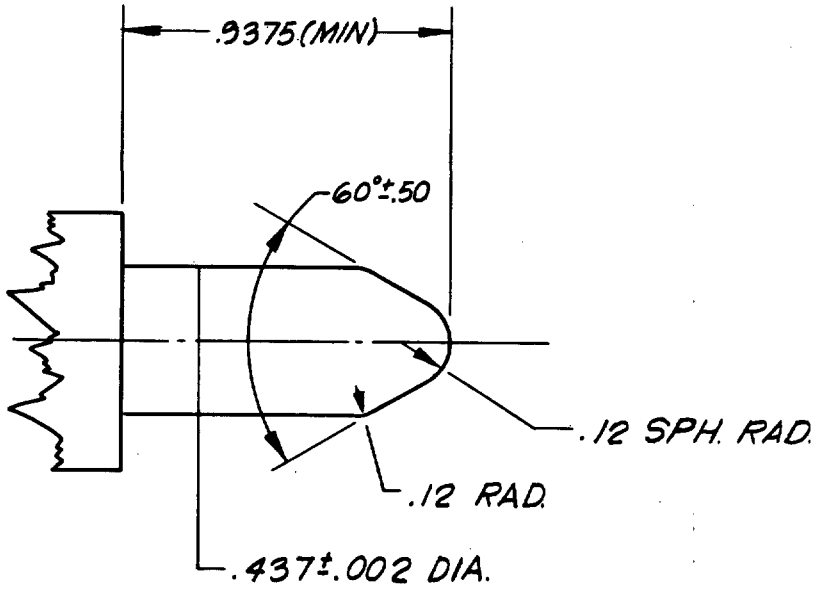
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DIMENSIONS IN INCHES
TOLERANCES ±.02
MAT'L STAINLESS STEEL
SPEC QQ-S-766

FIGURE 3. VENT NOZZLE

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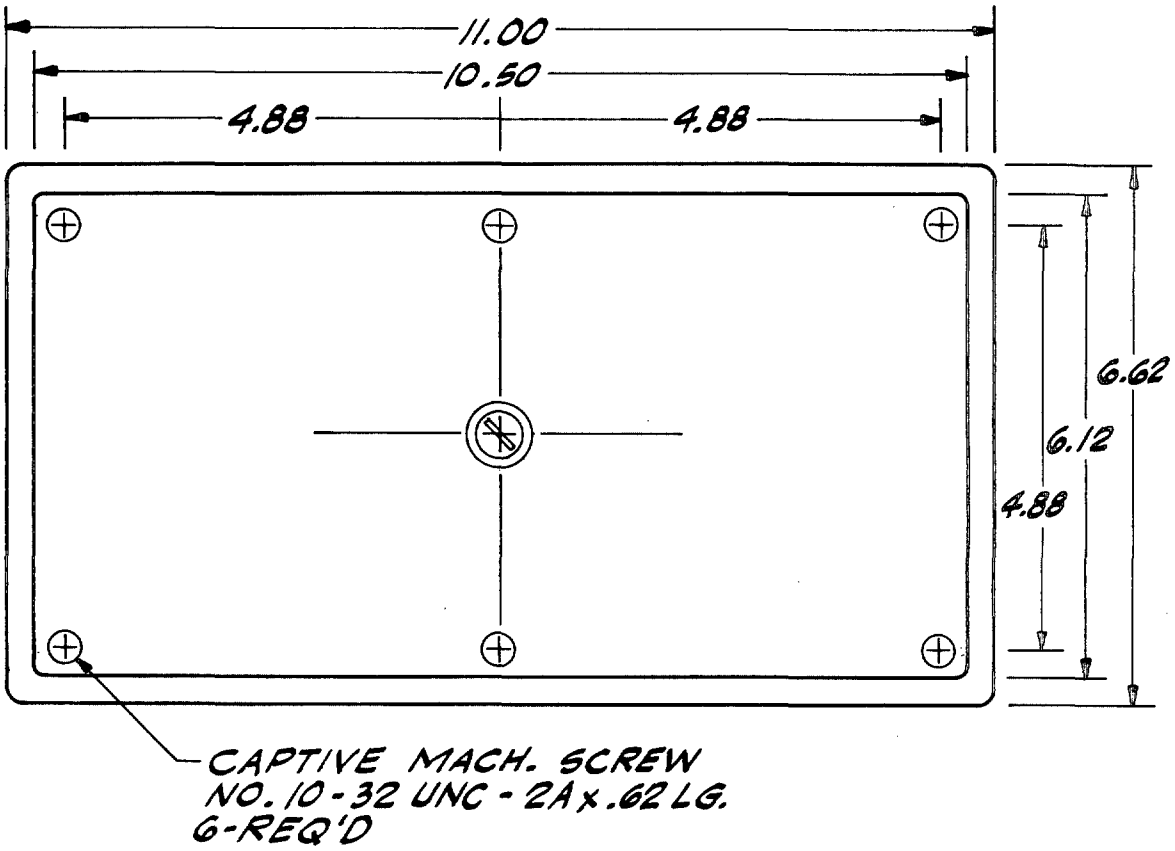


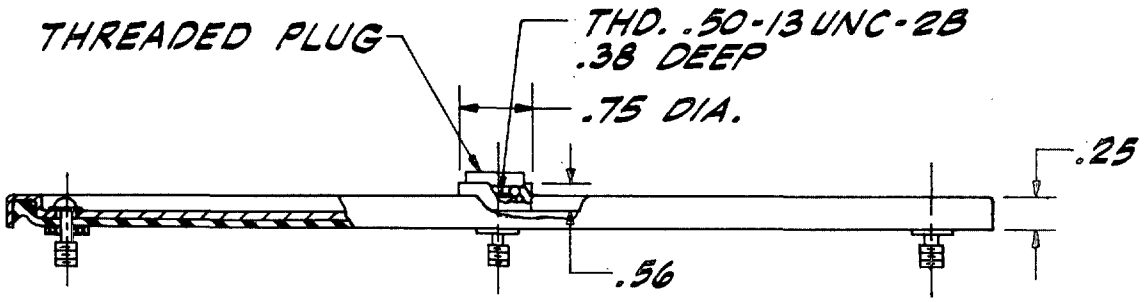
NOTE MATERIAL COPPER SPEC. QQ-C-502
EXCEPT CONDITION SHALL BE 3/4 HARD

ALL DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCES ARE AS FOLLOWS
DECIMALS ±.02

FIGURE 4. TERMINAL DETAIL

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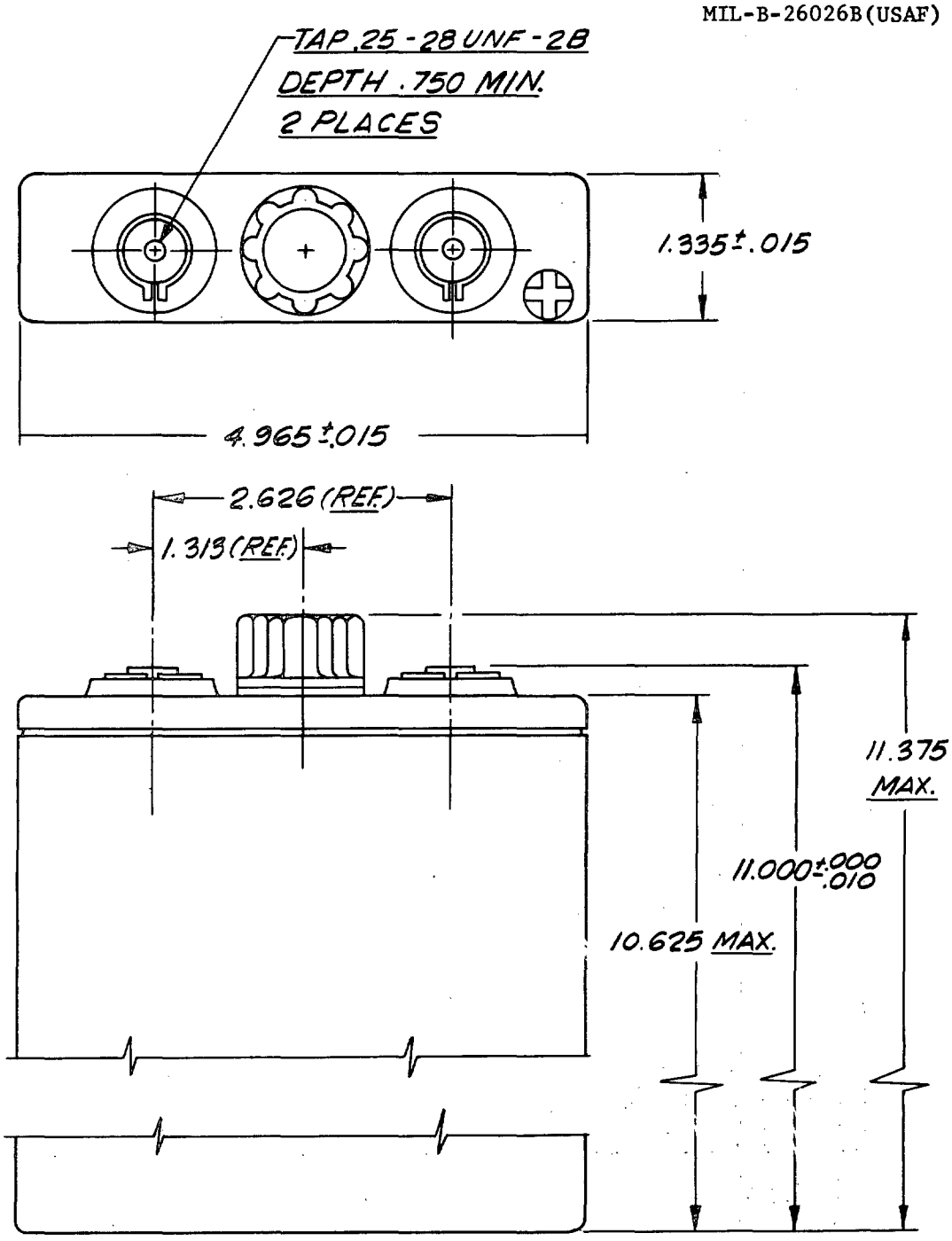




DIMENSIONS IN INCHES UNLESS
OTHERWISE SPECIFIED
TOLERANCE ±.03

FIGURE 5. BATTERY COVER

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DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED

FIGURE 6. CELL ASSEMBLY

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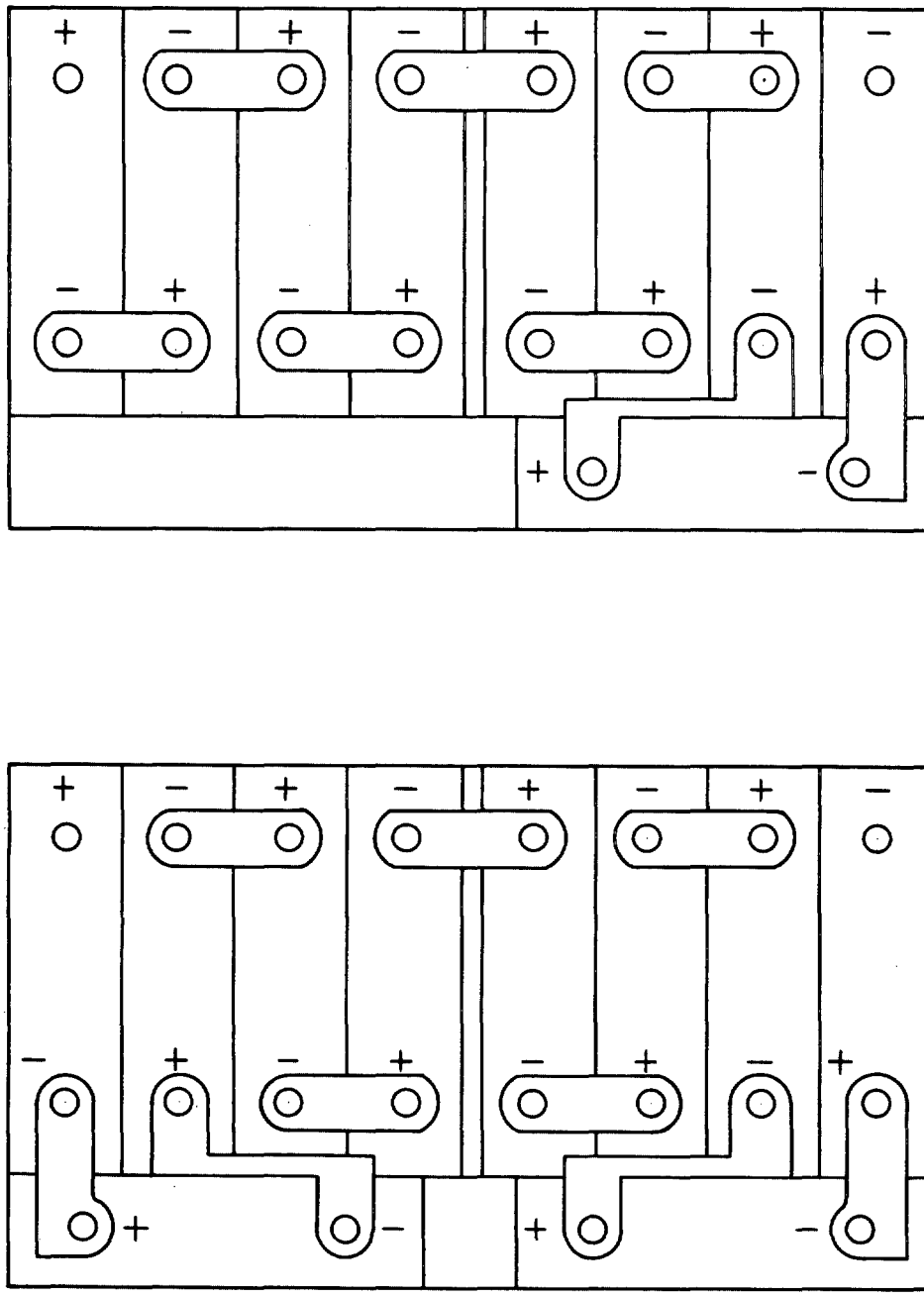


FIGURE 7. CELL ARRANGEMENT

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